

Tier 1 - Core ML

A. Regression Models

1) Linear Regression

x^i = input variable

y^i = output variable

w = weight

b = bias

$\hat{y}^i$ = predicted y (output)

Formula :

$$\hat{y}^i = wx^i + b$$

Cost Function = MSE $\sqrt{J}$

$$J = \frac{1}{2m} \sum_{i=1}^m L(y_i, \hat{y}_i)$$

$$L = (y_i - \hat{y}_i)^2$$

where, m = number of training examples

J = cost Function

L = Loss Function!

Note: Cost Function is calculated for overall dataset, and Loss Function is for individual training example.

Gradient Descent

This is what runs, or trains the model

Concept: Suppose you are standing on a curve, or a slope, if you want to reach the lowest point, you will:

1. Check where you are standing, is it uphill or downhill
2. Move forward or back in order to reach the lowest point

Slope/Curve = Loss Function

Uphill/Downhill = The differential

Gradient Descent will repeat the below steps, till it has minimized cost.

$$w = w - \alpha \frac{\partial L}{\partial w}$$

$$b = b - \alpha \frac{\partial L}{\partial b}$$

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Where, α = Learning rate

$$\frac{\partial L}{\partial w} = \frac{1}{m} \sum (\hat{y}_i - y_i) x_i$$

$$\frac{\partial L}{\partial b} = \frac{1}{m} \sum (\hat{y}_i - y_i)$$

Coding Optimizations

Libraries are most efficient, and are designed for matrix multiplications.

Calculations will look like:

$$\hat{y} = w \cdot x + b$$

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$$\frac{\partial L}{\partial w} = \frac{1}{m} [x^T \cdot (\hat{y} - y)]$$

Shapes:

$$x = (a, b)$$

{ $b \neq$ bias, here }

$$y = \hat{y} = (a, 1)$$

$$w = (b, 1)$$

$$\text{bias} = (1,)$$

Where in x , a = number of training examples

b = number of properties in each training example